Sequence-wide Moving-Support Discovery from Masked Colored Point Clouds

Stein particle transport, rigid coherence, and uncertainty-aware support inference

Technical research specification — 3 November 2026

Proposed mathematical design; not an experimentally validated algorithm

Abstract

Given a complete sequence of foreground-masked, calibrated RGB-D point clouds, infer which *initial persistent material points* belong to a support that undergoes physical motion. The first stage may internally estimate provisional free-point transport over the sequence, but its required outputs are per-point static/dynamic probabilities and a reliability-aware set of moving seeds. Subsequent stages recover rigid transformations, identify joint models, and use those models to compare stationary and multiple moving hypotheses for ambiguous points, especially those near hinges. Phase 1 investigates Stein variational gradient descent (SVGD) as a collective, appearance-guided transport mechanism. This document distinguishes the original SVGD theorem from an adaptation that treats physical surface points as particles, provides explicit score and kernel equations, derives a defensible sequence-level probabilistic observation model, and states the unresolved assumptions necessary for calibrated moving-support probabilities. No known part count, joint location, ground-truth motion, meshes, surfels, or input surface normals are assumed.

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1 Scope, deliverables, and non-goals

1.1 The precise question

Given observations $\mathcal{Q}_{0:T} = (\mathcal{Q}_0, \ldots, \mathcal{Q}_T)$, which initial material points *must be part of a moving rigid support* in order to explain the complete sequence? The requested probability is

$$\gamma_i = P(m_i = 1 \mid \mathcal{Q}_{0:T}), \qquad P_i^{\text{static}} = 1 - \gamma_i, \tag{1}$$

where $m_i = 1$ means that persistent point i belongs to a support that moves at some time during the sequence. This is an *ever-moving support membership* variable, not an instantaneous nonzero-velocity test. The original initialization index i is retained throughout.

1.2 Required outputs of each stage

Stage	Function	Output
1. Discovery	Internal appearance-guided transport; sequence-wide rigid-coherence and motion-evidence inference; conservative moving seeds	γ_i, uncertainty/calibration diagnostics, reliable moving-seed IDs
2. Rigid recovery	Estimate and separate rigid transformations from sufficiently reliable moving points	Candidate support motions $T_k(t) \in \mathrm{SE}(3)$
3. Joint inference	Compare joint families and optimize common joint geometry over time	Candidate articulated models and confidence
4. Support completion	Test unresolved points against static and each moving-model hypothesis with full-sequence observations	Refined assignment probabilities, including near-hinge points

Table 1: Separation of tasks. Stage 1 does not need to reconstruct all physical point trajectories accurately as a final product; it does need sufficiently informative provisional transport to discover moving seeds.

Non-goals for Stage 1: no GT part segmentation or axis; no predetermined number of moving parts; no compulsory exact point-to-point match; no hard decision that a weakly moving hinge-near point is static; no end-to-end joint fit; no inference based only on displacement amplitude; no declaration that SVGD's repulsion enforces physical rigidity.

2 Observation model and notation

2.1 Input point sets and masks

Time is indexed by $t = 0, \ldots, T$ with measured timestamps τ_t. The observed cloud for frame t is

$$\mathcal{Q}_t = \{\mathbf{z}_{j,t} = (\mathbf{y}_{j,t}, \mathbf{d}_{j,t})\}_{j=1}^{M_t}, \quad \mathbf{y}_{j,t} \in \mathbb{R}^3, \quad \mathbf{d}_{j,t} \in \mathbb{R}^3. \tag{2}$$

Only camera back-projections with *foreground mask equal to one and valid depth* enter $\mathcal{Q}_t$. $\mathbf{y}_{j,t}$ is the sample's *world-frame* XYZ position in meters; $\mathbf{d}_{j,t}$ is its measured RGB vector (a declared linear or gamma-encoded color convention must be fixed). M_t varies with time; raw samples have no persistent IDs. Retain the camera ID, camera center, and the original line of sight as *metadata* for visibility and uncertainty; all registration residuals operate on 3D points, not pixels.

Initialize persistent material hypotheses from the masked frame-0 cloud:

$$\mathcal{P}_0 = \{(\mathbf{x}_i^0, \mathbf{c}_i)\}_{i=1}^{N}, \qquad \mathbf{x}_i^0 \in \mathbb{R}^3, \quad \mathbf{c}_i \in \mathbb{R}^3. \tag{3}$$

N and the observed M_t need not coincide. A persistent point i has latent position $\mathbf{x}_{i,t}$, optional free velocity $\mathbf{v}_{i,t}$, and latent visibility $v_{i,t} \in \{0, 1\}$ at frame t. We write $\mathcal{X}_{0:T} = \{\mathbf{x}_{i,t}\}_{i,t}$ for all internal trajectories. To avoid confusion, $v_{i,t}$ denotes *visibility*, while bold $\mathbf{v}_{i,t}$ denotes *velocity*.

2.2 What the sequence does and does not measure

The observed clouds are independently re-sampled foreground surfaces. Their temporal changes combine: (i) physical material motion; (ii) small and often rapidly reversing XYZ depth fluctuations; (iii) RGB flicker, including correlated RGB channels; (iv) changing visibility; and (v) changed sampling density. A single observed $\mathbf{y}_{j,t}$ is *not* automatically a measurement of a particular material point $\mathbf{x}_i^0$. The archived visualizations also used display subsampling, so renderer flicker must not be identified with sensor noise without checking original samples.

For a stationary observed location with measurement error $\boldsymbol{\epsilon}_t$, consecutive increments satisfy

$$\Delta\mathbf{z}_t = \boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{t-1}, \qquad \mathrm{Cov}(\Delta\mathbf{z}_t, \Delta\mathbf{z}_{t+1}) = -R \tag{4}$$

for independent, identically distributed errors with covariance R. Negative correlation between successive increments therefore does *not* by itself establish a physical oscillation. Moreover, neither the sensor noise covariance nor the spatial density-bandwidth is uniquely determined by nearest-point distances.

2.3 Desired Stage-1 uncertainty semantics

γ_i must be interpreted as a *posterior probability under a declared model*, not as an arbitrary rescaling of a gradient norm or a rigid-fit residual. If the evidence model is only a heuristic/composite likelihood, its numerical outputs must be called *scores* until empirically calibrated against held-out data. Stage 1 may report high-confidence moving seeds and leave a large uncertain region; it is explicitly allowed to have low recall near joints.

3 General pipeline and information flow

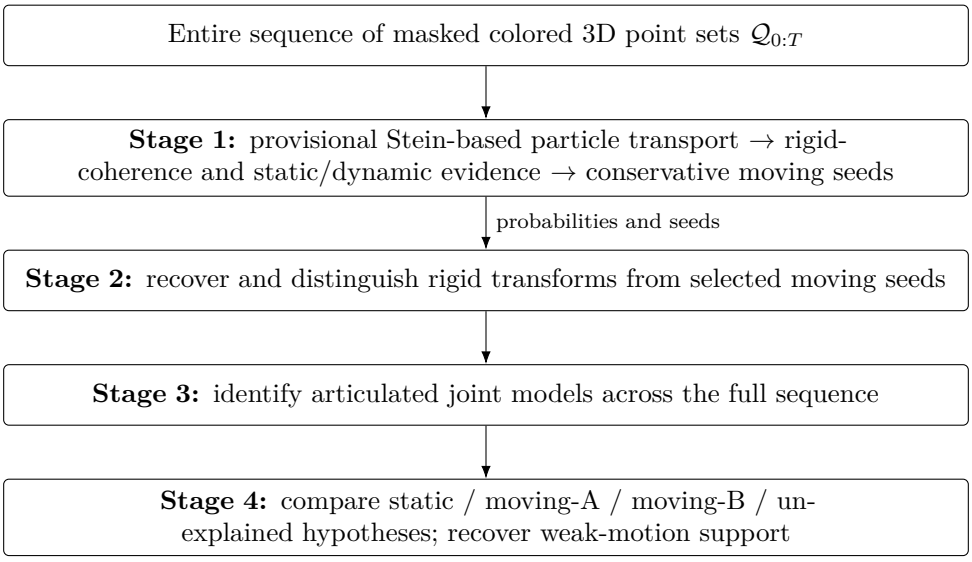


Figure 1: The four stages are conceptually separate. Motion, visibility, and candidate support relations inside Stage 1 are *latent inference variables*; only moving-support evidence is exported.

Stage 1 should select an evidentially strong subset. Stage 2 may then exploit a known effective rigid-recovery method. Stage 3 estimates joint type and geometry *from recovered rigid motions*. Stage 4 revisits ambiguous particles by comparing model-based predicted motion against the original observations; it does not use point displacement magnitude as a sufficient classifier.

4 What standard SVGD guarantees – and what it does not

4.1 The original distributional problem

Standard Stein variational gradient descent approximates a target density $p(\boldsymbol{\theta})$ by transporting samples $\boldsymbol{\theta}^1, \ldots, \boldsymbol{\theta}^L \in \mathbb{R}^d$ from an approximating distribution q [1]. The variational objective is $D_{\mathrm{KL}}(q\|p)$. For a small transport $T_\epsilon(\boldsymbol{\theta}) = \boldsymbol{\theta} + \epsilon\boldsymbol{\phi}(\boldsymbol{\theta})$,

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} D_{\mathrm{KL}}\big((T_\epsilon)_{\#}q\|p\big)\Big|_{\epsilon=0} = -\mathbb{E}_q\left[\nabla \log p(\boldsymbol{\theta})^\top \boldsymbol{\phi}(\boldsymbol{\theta}) + \nabla\cdot\boldsymbol{\phi}(\boldsymbol{\theta})\right]. \tag{5}$$

$T_{\#}q$ is the pushforward measure. In the vector-valued RKHS determined by a differentiable positive-definite kernel k, the optimal first-order decrease direction is proportional to

$$\boldsymbol{\phi}^*(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}'\sim q}\left[k(\boldsymbol{\theta}', \boldsymbol{\theta})\nabla_{\boldsymbol{\theta}'} \log p(\boldsymbol{\theta}') + \nabla_{\boldsymbol{\theta}'} k(\boldsymbol{\theta}', \boldsymbol{\theta})\right]. \tag{6}$$

The empirical update replaces the expectation by an average over L inference particles:

$$\boldsymbol{\theta}^{\ell,(k+1)} = \boldsymbol{\theta}^{\ell,(k)} + \eta_k \frac{1}{L}\sum_{r=1}^{L}\left[k(\boldsymbol{\theta}^{r,(k)}, \boldsymbol{\theta}^{\ell,(k)})\nabla_{\boldsymbol{\theta}^r} \log p(\boldsymbol{\theta}^{r,(k)}) + \nabla_{\boldsymbol{\theta}^r} k(\boldsymbol{\theta}^{r,(k)}, \boldsymbol{\theta}^{\ell,(k)})\right]. \tag{7}$$

k indexes *numerical iterations*, not physical frames, and η_k is a numerical step. The first summand transfers the target score to nearby particles; the second is the Stein repulsive interaction. This is a *local descent-direction* result under the paper's regularity assumptions, not a theorem that arbitrary steps or finite particles recover correct material-point trajectories.

4.2 The central interpretation distinction

Bayesian SVGD: Each $\boldsymbol{\theta}^\ell$ is one *hypothesis in unknown-parameter space*. For our full problem it could be a complete candidate trajectory configuration together with continuous nuisance variables, potentially of dimension $3N(T+1)$ or larger. Ordinary SVGD does not directly handle discrete labels m_i and z_{ij}. This is theoretically coherent but computationally formidable.

Physical-point Stein transport (research adaptation): Treat each stored material point $\mathbf{x}_i$ as a particle that is driven to follow the observed XYZ–RGB distribution. This is useful for generating provisional trajectories and collective corrections, but $\{\mathbf{x}_i\}$ are *not* posterior samples of the same unknown motion. Equation (7) cannot be claimed to deliver calibrated $P(m_i = 1 \mid \mathcal{Q}_{0:T})$ in this interpretation.

4.3 Original reference implementation

The cited repository [2], `python/svgd.py`, accepts an initial particle matrix and a callback returning $\nabla \log p$. It uses an RBF kernel and dense pairwise particle distances; its example is not an RGB-D registration system. Its Python file also uses Python-2-era print syntax. The public `update` routine exposes a bandwidth parameter but calls the kernel routine with the default median heuristic instead. At N material points, a full $N \times N$ dense matrix is $O(N^2)$ in memory and is unsuitable at approximately 10^5 points without localization or approximation. These are source-code observations, not claims about the quality of the SVGD theory.

5 Phase 1A – a differentiable XYZ–RGB target score

5.1 Why the point set alone is insufficient

The observed $\mathcal{Q}_t$ is discrete. Standard SVGD requires $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}, \mathbf{c})$ at a general query. The point samples do not themselves define a smooth 3D probability density or its gradient. A target model must be chosen and validated independently of the Stein transport kernel.

5.2 Explicit proposed target density

A simple, transparent *candidate model* is the observed Gaussian mixture

$$\widehat{p}_t(\mathbf{x}, \mathbf{c}) = \frac{1}{M_t} \sum_{j=1}^{M_t} \mathcal{N}_3(\mathbf{x}; \mathbf{y}_{j,t}, \Lambda^X_{j,t}) \mathcal{N}_3(\mathbf{c}; \mathbf{d}_{j,t}, \Lambda^C_{j,t}), \tag{8}$$

where $\mathcal{N}_3(\cdot; \boldsymbol{\mu}, \Sigma)$ is the *normalized* 3D Gaussian probability density; $\Lambda^X_{j,t} \succ 0$ and $\Lambda^C_{j,t} \succ 0$ are respectively the spatial and RGB *density-smoothing* covariance matrices (units m^2 and RGB-units^2). The factor $1/M_t$ is a normalization, not an unknown physical amplitude. This choice assumes equal contribution by each observed sample. With view-dependent sampling density or duplicated camera coverage, this assumption is biased and must be corrected at the *sampling-model* level, not by inventing unexplained per-point weights.

Important: Λ^X is not automatically the measured XYZ sensor covariance R^X. It also describes a chosen continuous representation of a sampled surface. Likewise Λ^C is not necessarily the true RGB flicker covariance R^C. Separating measurement uncertainty from representation smoothing is mandatory.

5.3 Derivation of the appearance-aware spatial score

Define the unnormalized mixture contribution for observed sample j:

$$a_{j,t}(\mathbf{x}, \mathbf{c}) = \mathcal{N}_3(\mathbf{x}; \mathbf{y}_{j,t}, \Lambda^X_{j,t}) \mathcal{N}_3(\mathbf{c}; \mathbf{d}_{j,t}, \Lambda^C_{j,t}), \quad r_{j,t}(\mathbf{x}, \mathbf{c}) = \frac{a_{j,t}(\mathbf{x}, \mathbf{c})}{\sum_{m=1}^{M_t} a_{m,t}(\mathbf{x}, \mathbf{c})}. \tag{9}$$

These $a_{j,t}$ are fully defined likelihood factors, *not arbitrary learnable point weights.* The $r_{j,t}$ are responsibilities only for computing a *density score*; they are not hard cross-frame material identities. Since

$$\nabla_{\mathbf{x}} \log \mathcal{N}_3(\mathbf{x}; \mathbf{y}, \Lambda^X) = (\Lambda^X)^{-1}(\mathbf{y} - \mathbf{x}), \tag{10}$$

we obtain exactly

$$\boxed{\mathbf{g}_t(\mathbf{x}, \mathbf{c}) := \nabla_{\mathbf{x}} \log \widehat{p}_t(\mathbf{x}, \mathbf{c}) = \sum_{j=1}^{M_t} r_{j,t}(\mathbf{x}, \mathbf{c})(\Lambda^X_{j,t})^{-1}(\mathbf{y}_{j,t} - \mathbf{x}).} \tag{11}$$

Thus a *3D spatial* force is shaped by both observed spatial proximity and RGB similarity. No target index is assigned to an initial material point. The number of observed samples M_t need not equal N.

A useful special case. If all $\Lambda^X_{j,t} = \sigma_X^2 I_3$, then

$$\mathbf{g}_t(\mathbf{x}, \mathbf{c}) = \sigma_X^{-2}\Big(\sum_j r_{j,t}(\mathbf{x}, \mathbf{c}) \mathbf{y}_{j,t} - \mathbf{x} \Big). \tag{12}$$

The score points to a local *colour-weighted barycenter*. It does not explicitly preserve material rigidity, and can be misleading at a moving/static overlap if the target responsibilities straddle two physical surfaces.

5.4 What varying colors and sample counts imply

The same surface may show RGB flicker and spatially correlated RGB-channel changes. A too-small Λ^C over-penalizes ordinary flicker; too-large Λ^C destroys the appearance information needed for motion. Similarly Λ^X must be large enough to give informative scores inside the expected transport basin yet small enough to distinguish nearby surfaces. A coarse-to-fine continuation is an *optimization strategy*, not proof of a particular physical noise model. Constant $1/M_t$ normalizes sample count but does not correct camera-overlap or point-density bias; comparing normalized empirical point distributions still requires compatible visibility/sampling support.

5.5 No independent material measurement follows from the score

$\widehat{p}_t(\mathbf{x}_i, \mathbf{c}_i)$ is a density of *observed samples* near the particle's location and color. It is *not* the probability that observed point j is material point i, and is not by itself a normalized likelihood $p(\mathcal{Q}_t \mid \mathcal{X}_t)$. A high score can be caused by arbitrary data-density gradients, not genuine object motion. We must not turn $\|\mathbf{g}_t\|$ into a moving/static probability.

6 Phase 1B – Stein transport of persistent physical particles

6.1 The transport candidate

For a fixed time t, let $\mathbf{x}_{i,t}^{(k)}$ be the provisional position of persistent material point i at numerical iteration k; $\mathbf{c}_i$ is its fixed reference appearance. Choose a smooth positive-definite interaction kernel

$$K_H(i,l) = \exp\Big[-\tfrac{1}{2}(\mathbf{x}_{i,t} - \mathbf{x}_{l,t})^\top H_X^{-1}(\mathbf{x}_{i,t} - \mathbf{x}_{l,t}) - \tfrac{1}{2}(\mathbf{c}_i - \mathbf{c}_l)^\top H_C^{-1}(\mathbf{c}_i - \mathbf{c}_l)\Big], \tag{13}$$

where $H_X \succ 0$ and $H_C \succ 0$ are transport-interaction bandwidth matrices, distinct from the observed target's Λ^X, Λ^C and from sensor-noise covariance. A simpler position-only K_H is an essential ablation. Since reference colours are fixed, derivatives below act only on XYZ coordinates.

For persistent point i, the proposed *position-only, Stein-inspired* field is

$$\boxed{\mathbf{\Phi}_{i,t} = \frac{1}{N}\sum_{l=1}^{N}\Big[K_H(l,i)\,\mathbf{g}_t(\mathbf{x}_{l,t}, \mathbf{c}_l) + \nabla_{\mathbf{x}_{l,t}} K_H(l,i)\Big], \qquad \mathbf{x}_{i,t}^{(k+1)} = \mathbf{x}_{i,t}^{(k)} + \eta_k \mathbf{\Phi}_{i,t}.} \tag{14}$$

For the Gaussian kernel,

$$\nabla_{\mathbf{x}_l} K_H(l,i) = K_H(l,i) H_X^{-1}(\mathbf{x}_i - \mathbf{x}_l). \tag{15}$$

In (14), target attraction transports density evidence; the kernel gradient introduces repulsion. Neither term imposes exact rigidity, preserves material trajectories, or supplies a calibrated probability.

6.2 Why the first-stage transport must be sequence-wide

Independent transport of frame t to $t+1$ can chase depth flicker, RGB flicker, and resampling, and may produce mutually inconsistent provisional trajectories. Let $\mathcal{X}_{0:T}$ be provisional point

trajectories. A generic temporal prior may be written

$$E_{\text{temp}}(\mathcal{X}_{0:T}) = -\log p_{\text{motion}}(\mathcal{X}_{0:T}), \tag{16}$$

with a declared free-motion model; e.g., a linear-Gaussian state model $\mathbf{s}_{i,t} = [\mathbf{x}_{i,t}^\top, \mathbf{v}_{i,t}^\top]^\top$ and covariance $P_{i,t}$, using actual $\Delta\tau_t$. A fixed-velocity prediction is a prior distribution, not a claim that every point moves at constant velocity.

There are *two different* ways to couple SVGD and temporal inference:

1. *Exact Bayesian formulation in principle:* Run standard SVGD over samples of the entire continuous trajectory posterior $p(\mathcal{X}_{0:T} \mid \mathcal{Q}_{0:T}, \text{other latents})$. One Stein particle is a *complete trajectory hypothesis*, so normal SVGD assumptions and interpretation apply. This is prohibitively high-dimensional for full dense clouds unless heavily structured and reduced.
2. *Practical Stage-1 research candidate:* Use (14) as a *provisional data-driven transport direction*, alternating with a separate temporal-prior optimization/smoother and coherence inference. This is an *adapted optimizer*, not standard SVGD on the sequence posterior. Its support probabilities must be derived and validated from a distinct probabilistic observation model.

The second option fits the user's desired simple particle-transport mechanism. It must be labeled honestly and compared against attraction-only, repulsion-only, and direct global distribution-gradient baselines.

6.3 Transport repulsion is not a physical constraint

Under (14), moving and stationary points that are spatially near may interact through K_H. At a static/dynamic boundary this can contaminate stationary support. A short interaction scale can reduce the contamination but may fail to transfer evidence across a moving part. A colour-aware kernel may help, but shared colours do not establish rigid membership. The interaction kernel must *not* be equated with a same-rigid-support probability. Later coherence inference is a separate step.

6.4 The computational issue

Full evaluation of K_H takes $O(N^2)$ entries; direct score evaluation from all M_t observed samples for all N transported points takes $O(NM_t)$. Sparse spatial neighborhoods, tree searches, low-rank kernel features, and chunked GPU reductions are candidate accelerations. Their approximation error must be measured against an exact small-scene oracle; large-scene runtime alone does not validate the method. The RBF kernel has global nonzero support, so sparsification changes the algorithm unless a compactly supported positive-definite kernel is chosen.

7 Phase 1C – a global distribution-level observation model

7.1 Why it is required

The Stein target score in (11) tells physical particles *where they can move* to increase local observed density. It does not, by itself, say how well a *whole transported cloud* explains a complete observation; neither does it define a Bayes factor between stationary and moving hypotheses. To infer γ_i we need an observation likelihood (or an explicitly uncalibrated surrogate).

7.2 An explicit finite-dimensional feature measurement

Let $\psi : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^D$ be a declared, differentiable feature map of XYZ and RGB; D is its feature dimension. For a kernel feature map, $\langle \psi(\mathbf{z}), \psi(\mathbf{z}') \rangle$ approximates or equals a selected positive-definite XYZ–RGB kernel. The current observed feature mean is

$$\mathbf{y}_t^F = \frac{1}{M_t} \sum_{j=1}^{M_t} \psi(\mathbf{y}_{j,t}, \mathbf{d}_{j,t}) \in \mathbb{R}^D. \tag{17}$$

For a *comparable visible subset* of transported material points $\mathcal{V}_t \subseteq \{1, \dots, N\}$, let

$$\mathbf{h}_t(\mathcal{X}_t, \mathcal{V}_t) = \frac{1}{|\mathcal{V}_t|} \sum_{i \in \mathcal{V}_t} \psi(\mathbf{x}_{i,t}, \mathbf{c}_i) \in \mathbb{R}^D. \tag{18}$$

Equation (18) is defined only for nonempty $\mathcal{V}_t$. $\mathcal{V}_t$ is a latent/estimated visible-material set, not a static/dynamic label. If visibility differs between observed and predicted distributions, the model must account for that difference explicitly; simply setting $\mathcal{V}_t = \{1, \dots, N\}$ can create false motion evidence.

A candidate *Gaussian feature-observation model* is

$$\mathbf{y}_t^F = \mathbf{h}_t(\mathcal{X}_t, \mathcal{V}_t) + \boldsymbol{\epsilon}_t^F, \qquad \boldsymbol{\epsilon}_t^F \sim \mathcal{N}_D(0, \Gamma_t), \tag{19}$$

where $\Gamma_t \succ 0$ describes feature uncertainty from measurement noise, temporal RGB variation, finite sampling and correlations. Its value must be estimated from data and the declared feature construction; it is not the raw RGB-D sensor covariance and not an arbitrary scalar.

The negative log-likelihood contribution (omitting a known normalization when Γ_t is fixed) is

$$E_{F,t} = \tfrac{1}{2} (\mathbf{y}_t^F - \mathbf{h}_t)^\top \Gamma_t^{-1} (\mathbf{y}_t^F - \mathbf{h}_t). \tag{20}$$

The derivative with respect to the position of a contributing persistent point $i \in \mathcal{V}_t$ is

$$\boxed{\nabla_{\mathbf{x}_{i,t}} \log p(\mathbf{y}_t^F \mid \mathcal{X}_t, \mathcal{V}_t) = \frac{1}{|\mathcal{V}_t|} J_\psi(\mathbf{x}_{i,t}, \mathbf{c}_i)^\top \Gamma_t^{-1} (\mathbf{y}_t^F - \mathbf{h}_t),} \tag{21}$$

where $J_\psi(\mathbf{x}, \mathbf{c}) = \partial \psi(\mathbf{x}, \mathbf{c}) / \partial \mathbf{x} \in \mathbb{R}^{D \times 3}$. This is a collective XYZ–RGB gradient with no point-to-point correspondence, and it can produce off-diagonal couplings between persistent points in a joint optimizer. It is *not* the same gradient as the KDE score in (11).

7.3 Relation to maximum mean discrepancy

For an exact RKHS kernel feature ψ, the squared norm of the difference of two mean embeddings is squared MMD [3]:

$$\mathrm{MMD}^2(\mathcal{P}_t, \mathcal{Q}_t) = \left\| \frac{1}{N} \sum_i \psi(\mathbf{x}_{i,t}, \mathbf{c}_i) - \frac{1}{M_t} \sum_j \psi(\mathbf{y}_{j,t}, \mathbf{d}_{j,t}) \right\|_{\mathcal{H}}^2. \tag{22}$$

For the finite feature model above, an isotropic Γ_t gives an MMD-like loss. A general calibrated Γ_t weights observable feature directions according to uncertainty. *MMD gradient descent is not SVGD*: MMD minimizes a discrepancy between two samples; SVGD targets a score-defined distribution through the Stein operator. They should be compared, not conflated.

7.4 This is a proposed feature likelihood, not a raw-cloud law

(19) defines the probability of a *summary statistic* $\mathbf{y}_t^F$ given a transported model. It does not automatically give the likelihood of every raw point, nor does equal mean-feature count guarantee material identity. Visibility, sample bias, feature approximation, and correlated observations can invalidate simplistic assumptions. A Bayes factor based on (19) is only calibrated if the feature noise model and model comparison are calibrated.

8 Phase 1D – visibility, re-sampling, and missing support

8.1 What the masks guarantee and what they do not

Masks guarantee that every raw observation entering $\mathcal{Q}_t$ is foreground-valid and that the persistent initialization uses only masked data. Masks do *not* guarantee that all physical material points are visible at every time. Neither a missing point nor a changed sample location proves motion. The scene can contain newly exposed foreground geometry absent from $\mathcal{P}_0$; Stage 1 must be able to call such evidence *unexplained* rather than forcing old particles toward it.

8.2 Visibility as a latent variable

For each point i and time t, $v_{i,t} = 1$ means an existing material point has an admissible measurement opportunity in the observed cloud. Define the full vector $V = (v_{i,t})_{i,t}$; it is *not* known from cloud cardinality. To evaluate line-of-sight visibility while remaining point-based, retain the calibrated 3D camera centers $\mathbf{o}_c$ and the original world-space 3D samples tagged by source camera. A nearer measured surface along the line from $\mathbf{o}_c$ toward $\mathbf{x}_{i,t}$ can occlude the deeper point. This uses geometry and sensor provenance, not pixel-residual tracking.

A useful generative formulation introduces $p(V \mid \mathcal{X}_{0:T}, \mathcal{C})$, where $\mathcal{C}$ denotes fixed camera calibration and observation metadata. If visibility cannot be established, either marginalize over visibility hypotheses or keep evidence uncertain. Do not set $v_{i,t} = 0$ whenever a current prediction falls outside the observed point support: the point may have moved outside the local search basin rather than become invisible. Likewise, no model may use a closer moving surface as a material observation of an occluded stationary point.

8.3 Unexplained observed geometry

Let B_t denote latent new/unrepresented observed support at time t. A complete sequence-level feature likelihood should distinguish explained existing support from B_t and observation noise. For Stage 1, B_t is a nuisance term: *no new persistent point IDs have to be exported.* That is intentional because point-birth and final support completion are not Stage-1 objectives. Forcing all observed feature mass to be explained by $\mathcal{P}_0$ will otherwise produce spurious transport toward newly exposed surfaces. The feature prediction in (18) is only a minimal illustration; a production likelihood must explicitly treat B_t and overlapping cameras.

8.4 Sampling-density and camera overlap

The raw empirical distribution is a distribution of *sensor samples*, not a uniform physical-area measure on the object. The $1/M_t$ factor removes a trivial multiplicative effect of sample count, not viewpoint-dependent density bias. A correct sampling model may involve each

observation's footprint on the visible surface, correlated samples from neighboring rays, or camera-stratified feature means. These quantities must be estimated from existing calibrated observations. Unjustified per-point weights should not be introduced merely to make a sum look convenient.

9 Phase 1E – sequence-wide rigidity and appearance coherence

9.1 Pairwise rigidity invariant

For two persistent points i, l on the same rigid physical support, their 3D distance is constant under any time-varying SE(3) motion:

$$\mathbf{x}_{i,t} = R_t \mathbf{x}_i^0 + \mathbf{b}_t, \qquad \mathbf{x}_{l,t} = R_t \mathbf{x}_l^0 + \mathbf{b}_t, \tag{23}$$

$$\|\mathbf{x}_{i,t} - \mathbf{x}_{l,t}\| = \|\mathbf{x}_i^0 - \mathbf{x}_l^0\|, \qquad R_t \in \mathrm{SO}(3),\ \mathbf{b}_t \in \mathbb{R}^3. \tag{24}$$

No pivot, hinge position, joint axis, or part center occurs in this condition. The distance derivative is

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\|\mathbf{x}_i - \mathbf{x}_l\|^2 = 2(\mathbf{x}_i - \mathbf{x}_l)^\top (\mathbf{v}_i - \mathbf{v}_l) = 0 \tag{25}$$

for shared instantaneous rigid motion. This is the differential rigid-coherence evidence from the latent free trajectories.

Define the distance-residual history for an initial-neighbor candidate edge (i, l):

$$e_{il,t} = \|\mathbf{x}_{i,t} - \mathbf{x}_{l,t}\| - \|\mathbf{x}_i^0 - \mathbf{x}_l^0\|, \qquad \mathbf{e}_{il} = [e_{il,1}, \ldots, e_{il,T}]^\top. \tag{26}$$

When points are close, $e_{il,t}$ inherits the correlated XYZ uncertainty of both endpoints and of the shared initialization. Let Ω_{il} denote its *temporal* covariance, including cross-time dependence. Under a local Gaussian approximation, the same-rigid-support hypothesis has $\mathbf{e}_{il} \sim \mathcal{N}_T(0, \Omega_{il})$. This is a conditional likelihood, not a complete calibrated posterior.

9.2 Why rigid invariance is necessary but not sufficient

Constant distance is also preserved for pairs of static points; different supports may accidentally preserve a particular pairwise distance (e.g., a stationary point on a rotation axis). A sparse graph with all edge lengths preserved need not be globally rigid. The algorithm should use redundant geometric relationships and complete-sequence evidence; it must not classify a pair as shared support solely because $\mathrm{Var}_t\, d_{il,t}$ is small.

9.3 Relative appearance

Reference colours $\mathbf{c}_i$ and $\mathbf{c}_l$ remain attached to persistent material hypotheses. Their relative appearance $\Delta\mathbf{c}_{il} = \mathbf{c}_i - \mathbf{c}_l$ is unchanged by transport *by construction*, so it cannot independently *prove* correct motion. The informative quantity is whether the *transported spatial arrangement of colours* predicts the observed XYZ–RGB sequence. In particular, comparing $\mathbf{c}_i$ with unrelated current $\mathbf{d}_{j,t}$ without a distributional model can produce false support evidence. Relative colour can help define candidate neighborhoods, especially if global illumination perturbations partly cancel, but colour similarity alone does not identify rigid membership.

9.4 Latent support graph

Let E be a candidate graph of initial persistent-point relationships, created from initial *3D* neighborhoods. The graph must include enough edges to detect changes and not assume that every nearby point belongs to one part. Define

$$z_{il} = \mathbb{I}\{\text{points } i, l \text{ belong to one rigid support}\}, \qquad w_{il} = P(z_{il} = 1 \mid \mathcal{Q}_{0:T}). \tag{27}$$

Near physical contact, graph edges may cross rigid-body boundaries. Their uncertainty is the purpose of inference. Forming an edge prior based on initial spatial proximity is not equivalent to fixing $z_{il} = 1$. The full posterior over $Z = (z_{il})_{(i,l)\in E}$ must respect consistency among edges; independent edge Bernoulli posteriors do not automatically form a valid rigid partition.

9.5 Hinge-near motion

If a body rotates by angle θ_t around an unknown axis, a point at perpendicular radius r_i has displacement magnitude

$$\|\mathbf{x}_{i,t} - \mathbf{x}_i^0\| = 2r_i|\sin(\theta_t/2)| \approx r_i|\theta_t| \quad (|\theta_t| \ll 1). \tag{28}$$

For a 1-degree rotation, a point at 1 cm radius moves about 0.175 mm, whereas one at 20 cm moves about 3.49 mm. The first displacement may be smaller than measured RGB-D jitter, although both are on the same rigid support. Stage 1 is *not required* to classify the first point with confidence. After later rigid/joint recovery, Stage 4 can test this point under the recovered constrained motion hypothesis.

10 Phase 1F – static/dynamic evidence and probability

10.1 The latent variables and posterior target

Let $m_i \in \{0, 1\}$ be an *ever-moving support* indicator, $\mathbf{m} = (m_i)_{i=1}^N$, and Z be the unknown same-support relations. Let V denote visibility, B denote unexplained/newly visible support, and $\mathcal{X}_{0:T}$ the provisional free-point trajectories. The full conceptual Bayesian problem is

$$\boxed{p(\mathcal{X}_{0:T}, \mathbf{m}, Z, V, B \mid \mathcal{Q}_{0:T}) \propto p(\mathcal{Q}_{0:T} \mid \mathcal{X}_{0:T}, V, B)\, p(\mathcal{X}_{0:T} \mid \mathbf{m}, Z)\, p(V, B, \mathbf{m}, Z).} \tag{29}$$

All factors must be specified to claim numerical posterior probabilities. In particular, the observation term must distinguish jitter and flicker from real changes, and the trajectory prior must distinguish stationary, arbitrary free motion, and shared rigid motion. This factorization is an inference target, not a closed-form solution.

The desired marginal is

$$\boxed{\gamma_i = \sum_{\mathbf{m}_{-i}} \sum_{Z} \sum_{V} \sum_{B} \int p(\mathcal{X}_{0:T}, m_i = 1, \mathbf{m}_{-i}, Z, V, B \mid \mathcal{Q}_{0:T})\, \mathrm{d}\mathcal{X}_{0:T}.} \tag{30}$$

In practice, discrete summations and the high-dimensional integral require approximation. We must report which approximation is used, rather than call a sigmoid of residual energy a Bayesian posterior.

10.2 Why model evidence, not best residual, discriminates motion

For a candidate support S, compare two *sequence* hypotheses:

$$H_S^0 : \mathbf{x}_{i,t} = \mathbf{x}_i^0 \quad \text{for } i \in S \text{ (up to latent initial geometry uncertainty)}, \tag{31}$$

$$H_S^1 : \mathbf{x}_{i,t} \text{ follows a coherent but unknown motion trajectory for } i \in S. \tag{32}$$

Both models include the same observation noise and visibility uncertainty; H_S^1 additionally has a motion prior and extra latent degrees of freedom. For $h \in \{0, 1\}$, define the integrated evidence

$$L_S^h = p(\mathbf{y}_{0:T}^F \mid H_S^h, \text{rest of scene}) = \int p(\mathbf{y}_{0:T}^F \mid \Theta_S, H_S^h)\, p(\Theta_S \mid H_S^h)\, \mathrm{d}\Theta_S, \tag{33}$$

where Θ_S collects unknown trajectories, visibility, and any other continuous nuisance parameters for S. Priors and scene conditioning must be declared. The normalizing/marginalizing integral is precisely what penalizes overly flexible moving explanations.

Given the prior $\pi_S = P(H_S^1)$, Bayes' rule yields

$$P(H_S^1 \mid \mathbf{y}_{0:T}^F) = \frac{\pi_S L_S^1}{\pi_S L_S^1 + (1 - \pi_S) L_S^0}. \tag{34}$$

This support-level quantity can be propagated to particles through uncertain support membership. No guarantee of calibrated probabilities follows from substituting a best-fit cost for L_S^h; evidence approximations such as Laplace, importance sampling, or structured variational inference require explicit approximations and checks.

10.3 How moving seeds are selected

An initial point gets reliable moving evidence only when the observed sequence is inconsistent with its plausible static explanation *and* consistent with coherent nontrivial motion of a support. A point showing large but isolated and nonrepeatable residuals near an occlusion is not a reliable moving seed. Conversely, points close to the hinge may remain uncertain and are deferred.

A seed set $\mathcal{S}_{\text{seed}}$ may be chosen by a *declared posterior decision criterion*, e.g., an expected false-discovery budget: for selected set A, the posterior expected number of false moving selections is $\sum_{i \in A}(1 - \gamma_i)$. If the γ_i values are not calibrated, such a bound is not statistically justified; first calibrate on held-out synthetic/real controls or report scores without probability wording. No arbitrary displacement/velocity gate is required in the inference definition.

10.4 Shared-motion hypotheses at the Stage-1 boundary

One must not prematurely demand final rigid recovery in Stage 1. Provisional rigidity is an *evidence test*: trajectories of candidate neighbors should be compatible with common rigid motion, and competing static/dynamic explanations should have different sequence evidences. Stage 2 remains responsible for accurate final SE(3) transformations. If Stage 1 exports provisional trajectories for debugging, they are auxiliary diagnostics, not the promised final output.

10.5 A concrete whole-sequence objective for implementation

To make the relation between the probabilistic target and an optimizer explicit, one possible *structured approximation* to the negative log posterior is

$$\begin{aligned}\mathcal{E}(\mathcal{X}_{0:T}, \mathbf{m}, Z, V, B) = &\sum_{t=0}^{T} E_{F,t}(\mathcal{X}_t, V_t, B_t) - \log p_{\text{motion}}(\mathcal{X}_{0:T} \mid \mathbf{m}) \\ &+ \frac{1}{2} \sum_{(i,l)\in E} z_{il}\, \mathbf{e}_{il}^{\top} \Omega_{il}^{-1} \mathbf{e}_{il} - \log p(\mathbf{m}, Z, V, B) + C_{\text{norm}},\end{aligned} \tag{35}$$

where $E_{F,t}$ is the normalized distribution-feature observation cost from (20); V_t and B_t determine which observed and predicted support is comparable; p_{motion} is a declared, *label-conditioned* temporal prior; z_{il} and $\mathbf{e}_{il}$ are the rigid-support indicator and distance history; and C_{norm} collects normalization terms, which may *depend* on labels and model complexity. Dropping the latter is acceptable for some conditional trajectory optimizations, but *not* generally for Bayes-factor or support-probability evaluation.

The motion prior must compare stationarity against a broad but physically justified moving process: for example, an explicit static trajectory component plus a moving trajectory component with estimated temporal process covariance. A generic "moving means a free position at every frame" model would overfit jitter. The rigid-coherence term in (35) is a Gaussian *approximation* to the edge likelihood; it is not automatically a valid normalized joint prior on a rigid partition, because edge residuals share points and edges.

A useful differential check is the Jacobian of the pairwise distance residual. Define the unit relative vector

$$\mathbf{n}_{il,t} = \frac{\mathbf{x}_{i,t} - \mathbf{x}_{l,t}}{\|\mathbf{x}_{i,t} - \mathbf{x}_{l,t}\|} \tag{36}$$

when the denominator is nonzero. Under small perturbations,

$$\delta e_{il,t} \simeq \mathbf{n}_{il,t}^{\top}(\delta\mathbf{x}_{i,t} - \delta\mathbf{x}_{l,t}) - \mathbf{n}_{il,0}^{\top}(\delta\mathbf{x}_i^0 - \delta\mathbf{x}_l^0). \tag{37}$$

If ζ stacks the uncertain reference and time-varying states, J_{il} stacks the derivatives in (37), and P_ζ is their *joint* covariance, then

$$\Omega_{il} \simeq J_{il} P_\zeta J_{il}^{\top}. \tag{38}$$

In particular, repeated use of the frame-0 reference creates nonzero temporal cross-covariance; treating all $e_{il,t}$ as independent overstates rigidity evidence. Degenerate pairs whose separation is indistinguishable from point-position uncertainty should not be assigned an artificially precise direction.

Equation (35) defines what a candidate implementation seeks to explain. The physical-point Stein update (14) is a *proposal mechanism* for exploring its trajectory variables; it is not generally the exact gradient of (35). Any accepted update must be checked against an objective compatible with the same observation and prior models. If exact posterior inference remains infeasible, report approximate evidence scores and calibrate the resulting seed decisions explicitly.

11 Phase 1G – an implementable research algorithm

11.1 Alternating, sequence-wide scheme

The following is a *proposed approximation* to (29), not an exact SVGD posterior sampler. The key design is to keep the high-dimensional trajectories latent and only export evidence probabilities/seeds.

1. **Mask and initialize.** Construct $\mathcal{Q}_t$ from valid foreground XYZ–RGB for every frame. Preserve original point ID, reference RGB, camera provenance and measurement covariance. Initialize persistent points from $\mathcal{Q}_0$ without guessing static/dynamic labels.

2. **Estimate observation variability.** Fit a model for stable-region XYZ jitter, RGB flicker, cross-channel correlation and temporal correlation. Distinguish discrete depth quantization and camera-sampling fluctuation from genuine motion. Independently define and validate KDE bandwidths Λ^X, Λ^C.

3. **Build differentiable current-frame targets.** Construct $\widehat{p}_t$ and its 3D XYZ score $\mathbf{g}_t$ from (8)–(11), using only currently valid masked 3D samples. Do not equate it with a material-point observation likelihood.

4. **Initialize provisional trajectories.** Use a declared free-motion prior over the complete sequence, with explicit uncertainty. Do not mark unknown velocities as observed zero velocity.

5. **Stein-inspired point transport.** For each frame, compute attraction and kernel interaction from (14). Use physically meaningful 3D support and a step strategy that checks the true sequence objective; store candidate displacements, not fake measurements.

6. **Whole-sequence temporal refinement.** Update provisional trajectories using the declared motion prior and the global colored-cloud feature likelihood, including correlated measurement covariance, visibility and unexplained observation support. Reevaluate data agreement over the entire sequence rather than treating each pair as independent.

7. **Rigid coherence.** Build candidate initial 3D neighborhood edges and evaluate their distance histories (26), with uncertainty. Infer candidate same-support relations without assuming the number of rigid parts or any known pivot. Distinguish static supports from rigidly moving supports via world-frame motion evidence.

8. **Evaluate competing evidence.** Compare stationary and coherent-motion hypotheses over the full sequence using (33), with declared priors and sufficient control for model flexibility. Iterate transport/coherence/probability updates while checking out-of-sample agreement to avoid self-reinforcing errors.

9. **Export and stop.** Output γ_i or clearly labeled uncalibrated moving scores, confidence diagnostics, and reliable moving seeds. Do not force a binary label on every point or produce final SE(3)/joint estimates here.

11.2 The exact responsibilities of the three numerical operators

Operator	Mathematical role	Not allowed to claim
KDE target score $\mathbf{g}_t$	Local appearance-conditioned 3D density gradient	Calibrated moving probability; correct material correspondence
Stein field $\mathbf{\Phi}_{i,t}$	Collective provisional particle displacement, with repulsion	Rigid-body constraint; correct point identity
Feature likelihood / evidence	Quantifies observed full-cloud explanatory power, conditional on visibility and noise assumptions	Calibration without validated Γ_t, priors, and sampling model

11.3 Optimization and uncertainty caveats

- Transport under large displacement can leave the informative gradient basin of a narrow density. Assess coarse-to-fine continuation on synthetic oracle cases, but do not confuse its chosen scale with physical noise.

- If the objective includes reestimated visibility/point support, "the loss decreased" is meaningful only when compared against one consistent, normalized likelihood with the same observed evidence. Dropping hard measurements during a step is not objective improvement.
- Raw repeated RGB-D samples from adjacent cameras and times are not all independent; treating them as independent gives false posterior certainty.
- No arbitrary random initialization of a "moving" subset is permitted; if multiple candidate explanations remain possible, preserve hypotheses or uncertainty.
- SVGD repulsion can alter sampling density and drive false deformation; compare it explicitly with target-score-only transport.
- Recoverable motion may be low amplitude even with genuine articulation. Stage 1 values precision of moving seeds over exhaustive support recall.

11.4 Computational interfaces

The minimal conceptual interface is:

Input	Masked $\mathcal{Q}_{0:T}$, original 3D initialization $\mathcal{P}_0$, calibration metadata, time stamps, observation uncertainty model
Latent internal state	Provisional $\mathcal{X}_{0:T}$; target scores; visibility; uncertain support graph and labels; scenario evidences
Output	$(\gamma_i, 1-\gamma_i)$ or uncalibrated evidence score, seed IDs, uncertainty/diagnostic report
Not output	Final rigid trajectories, joint axis, part count imposed a priori, completed near-hinge support

12 Remaining stages – from moving seeds to complete articulation

12.1 Stage 2: accurate rigid-motion recovery

Use reliably moving seeds, potentially forming more than one coherent support, to recover one or several rigid transformations over time. For a candidate support S_k with initial positions $\mathbf{x}_i^0$ and estimated tracked positions or observation-compatible positions $\widetilde{\mathbf{x}}_{i,t}$, a weighted rigid fit is

$$(\widehat{R}_{k,t}, \widehat{\mathbf{b}}_{k,t}) = \underset{R\in\mathrm{SO}(3),\mathbf{b}\in\mathbb{R}^3}{\arg\min} \sum_{i\in S_k} \omega_{i,t} \|\widetilde{\mathbf{x}}_{i,t} - (R\mathbf{x}_i^0 + \mathbf{b})\|^2. \tag{39}$$

$\omega_{i,t}$ are fit weights derived from the Stage-1 support evidence and observation uncertainty; no GT groups or axes are used. If complete material correspondences are unavailable, an existing rigid-reconstruction method can evaluate a full-cloud colored-distribution likelihood instead of a least-squares correspondence fit; equation (39) simply states the constrained rigid target. Distinct candidate supports should be separated using consistency of the estimated transforms, not a prespecified count K.

12.2 Stage 3: joint-model identification

Using the recovered transforms $\widehat{T}_{k,t} \in \mathrm{SE}(3)$, estimate whether their *relative* motion is best described by a revolute, prismatic, fixed, or other supported kinematic family. A general joint model $\mathcal{J}$ with parameters $\boldsymbol{\alpha}$ and time-varying joint coordinate q_t predicts

$$T_{\mathrm{rel}}(t) \simeq T_{\mathrm{rel}}(q_t; \boldsymbol{\alpha}, \mathcal{J}). \tag{40}$$

The joint origin, direction, and axis are *estimated*, never supplied. Compare models with identical held-out motion/measurement evidence or an explicit complexity-aware model comparison; the model with the smallest unconstrained training residual is not automatically best.

12.3 Stage 4: recover the uncertain support

For any unresolved original point i, compare hypotheses

$$H_i^S : \mathbf{x}_{i,t} = \mathbf{x}_i^0, \tag{41}$$

$$H_i^k : \mathbf{x}_{i,t} = R_{k,t}\mathbf{x}_i^0 + \mathbf{b}_{k,t}, \quad k \in \{1, \dots, K\}, \tag{42}$$

$$H_i^U : \text{not adequately explained by the current candidate models.} \tag{43}$$

H_i^U is essential so a point is not forced into an incorrect part merely because one available model is marginally better. The transform $T_{k,t} = (R_{k,t}, \mathbf{b}_{k,t})$ is obtained from the recovered and joint-optimized model. Use the *full original XYZ–RGB observations* and visibility/uncertainty when evaluating each hypothesis. With priors π_i^h and valid evidences L_i^h,

$$P(H_i^h \mid \mathcal{Q}_{0:T}) = \frac{\pi_i^h L_i^h}{\sum_{g \in \{S,1,\dots,K,U\}} \pi_i^g L_i^g}, \qquad P_i^{\text{dynamic}} = \sum_{k=1}^{K} P(H_i^k \mid \mathcal{Q}_{0:T}). \tag{44}$$

The posterior is conditioned on recovered model uncertainty; treating the estimated transforms as exact would overstate confidence. Carefully separate fitting data and evidence used to assess model generalization (or use joint inference) so the same noisy samples are not counted twice as independent confirmation.

Why the hinge boundary is recoverable later. In a revolute part, points near the axis can move below per-frame XYZ jitter. Their entire expected trajectories under $T_{k,t}$ are nevertheless determined by the joint model inferred from points with stronger displacement. A near-axis point is assigned by comparing the likelihood of (41) under stationary and moving hypotheses, *not* by requiring its individually observed motion to exceed a threshold. If two hypotheses produce identical observable distributions for a point throughout the sequence, its membership is genuinely unidentifiable without additional evidence; report uncertainty.

13 Validation program and research decisions

13.1 Required controls before declaring Phase 1 successful

1. **Noise-only static control:** repeated frame-varying XYZ/RGB observations of a stationary surface, including depth quantization and correlated RGB flicker. Evaluate false moving-seed rate and probability calibration, not just cloud registration error.
2. **Rigid low-amplitude control:** synthetic rotation with known geometry; test near-axis points separately from more observable exterior points. Stage 1 is allowed to defer the former, but must not contaminate the seed set with static neighbors.
3. **Large displacement control:** moving colored geometry with a shifted support exceeding narrow KDE width. Verify whether the score provides an optimization basin and whether coarse-to-fine continuation actually helps.
4. **Occlusion and disocclusion control:** moving foreground surface crossing a static surface; ensure occluded static particles are not moved toward the front surface, and newly exposed geometry is not forced onto old point IDs.

5. **Appearance controls:** real and synthetic RGB flicker; uniform-colour surfaces; repeated-color textures; shared illumination drift; compare geometry-only, colour-only, and joint scores.
6. **Multiple moving supports:** two moving parts and one stationary part; check that a single Stein interaction field does not merge distinct motions at an overlap.

13.2 Suggested scene selection

Use one synthetic and one real scene for initial diagnostics, then expand. Synthetic examples include `blade_103706` (prismatic), `USB_100109` (rotational), and `scissor_11100`. Real controls include the supplied fridge-plastic and white-drawer RGB-D sequences. Camera calibration and frame metadata may be used at inference time; ground-truth moving labels, joints, or transforms are reserved for evaluation.

13.3 Ablations needed to attribute gains correctly

Ablation	Question answered
No transport / stationary	How much evidence arises solely from measurement jitter and changing sample support?
Target-score attraction only	Does the observed XYZ–RGB score move particles adequately without Stein repulsion?
Full Stein update	Does the repulsion/smoothing term improve high-precision moving seeds or distort shape?
Geometry vs appearance	Does RGB add useful motion evidence despite flicker, or mainly increase false positives?
Pairwise frame transport vs sequence-wide	Does full-sequence inference suppress jitter and retain large articulated motion?
Without vs with rigidity coherence	Does support coherence distinguish meaningful motion and protect static boundaries?
With vs without visibility treatment	How much static drift is attributable to occluded points being pulled toward other surfaces?
Feature likelihood vs KDE fit score	Are transport quality and probability evidence genuinely different?

13.4 Primary quantitative metrics

On synthetic scenes with GT used *only after inference*, report moving-seed precision, moving-seed recall, expected calibration error or reliability diagram for γ_i, ranking AUPRC under class imbalance, calibration conditioned on hinge radius and occlusions, and false moving evidence for stable regions. Report trajectory and cloud-distribution errors *as internal diagnostics*, not as Stage-1 success criteria. Evaluate Stage 4 separately on near-hinge recovery after recovered rigid/joint models. On real scenes, use stable intervals and consistency/held-out evidence; avoid inventing per-point GT labels unavailable from the measurements.

13.5 Failure criteria

A method fails as a motion-support detector if it: (i) follows high-frequency noise into false moving seeds; (ii) declares all moving support static because the empirical target is dominated by static points; (iii) pushes hidden stationary points onto occluding surfaces; (iv) merges independent moving parts via kernel coupling; (v) assigns high moving probability solely from large RGB flicker; or (vi) advertises numerical posterior probabilities without a calibrated likelihood and prior. A low distributional loss alone is not success.

14 Unresolved mathematical choices and their consequences

Choice	Why it must be specified	Failure if arbitrary
Target Λ^X	Continuous support of sampled 3D surfaces vs actual depth noise	No gradient at large motion; merging nearby surfaces
Target Λ^C	RGB flicker and reference appearance uncertainty	Colour-driven false motion; ignoring useful texture
Interaction H_X, H_C	How transport information propagates among physical particles	Boundary contamination; no coherent propagation
Feature map ψ	Which XYZ–RGB distribution differences are observable	Failure to distinguish moving geometries with similar summaries
Feature covariance Γ_t	Correlated measurement/sampling errors in collective features	Uncalibrated evidence and false probability certainty
Visibility V	Occlusion and unsupported old points	Pulling static points to moving occluders
Unexplained support B	Newly observed geometry absent from initialization	Artificial transport to fill disocclusion
Free-motion prior	Distinguish sustained trajectories from jitter without GT rigidity	Every flicker accepted as motion or genuine motion suppressed
Rigidity graph E, Z	Which candidate pairs share rigid motion, with uncertainty	False rigid groups; hinge confusion
Model complexity / evidence	Why extra motion degrees of freedom are not rewarded automatically	Moving explanation always wins training residual
SVGD particle semantics	Physical points vs posterior-hypothesis samples	Invalid Bayesian claims
Sampling correction	Unordered clouds do not uniformly sample physical area	Dense camera regions dominate probability
Probability calibration	Convert support evidence into interpretable γ_i	Arbitrary scores mislabeled as probabilities

15 Notation glossary

Symbol	Meaning
t, T, τ_t	Frame index, final frame index, and physical timestamp
$\mathcal{Q}_t, M_t$	Current masked colored observed 3D cloud and its number of samples
$\mathbf{y}_{j,t}, \mathbf{d}_{j,t}$	Observed XYZ and RGB of raw point j at frame t
$\mathcal{P}_0, N$	Reference persistent colored cloud and number of persistent points
$\mathbf{x}_i^0, \mathbf{c}_i$	Initial XYZ and attached reference RGB for persistent material ID i
$\mathbf{x}_{i,t}, \mathbf{v}_{i,t}$	Latent transported XYZ and free 3D velocity of persistent point i
$\mathcal{X}_{0:T}$	Entire set of latent provisional trajectories across all points/frames
$v_{i,t}, V$	Latent visibility indicator and collection thereof (not velocity)
B_t, B	Unrepresented/newly exposed observed geometric support and collection
$\widehat{p}_t$	Candidate continuous observed XYZ–RGB density
Λ^X, Λ^C	Spatial and RGB *target-density bandwidth covariances*

Symbol	Meaning
$a_{j,t}, r_{j,t}$	Explicit Gaussian mixture contribution and normalized score responsibility
$\mathbf{g}_t$	Spatial gradient of observed log density at a colored candidate
K_H, H_X, H_C	Stein physical-particle interaction kernel and bandwidth covariances
$\mathbf{\Phi}_{i,t}, \eta_k$	Provisional particle-transport field and numerical optimization step size
$\psi, D, \mathbf{y}_t^F$	XYZ–RGB feature map, feature dimension, and observed feature mean
$\mathbf{h}_t, \Gamma_t$	Predicted visible feature mean and its observation-error covariance
Z, z_{il}, w_{il}	Same-support graph, binary edge membership, and edge posterior
m_i, γ_i	Ever-moving support label and marginal moving probability
Ω_{il}	Full temporal covariance of relative-distance residual history
$R_t, \mathbf{b}_t$	Rigid rotation and translation for a candidate support
H_S^0, H_S^1, L_S^h	Stationary/moving support hypotheses and model evidences
$\mathcal{J}, \boldsymbol{\alpha}, q_t$	Joint family, estimated joint geometry, and time-varying joint coordinate

16 Conclusions and decision gate

A defensible Phase 1 cannot be reduced to an independent displacement test, nor can it assume that every colored 3D sample remains the same material point. It should use full-sequence XYZ–RGB distribution information to generate provisional transport, search for invariant relative geometric structure, and compare stationary versus coherent moving explanations under actual observation uncertainty. Stein particle transport is one plausible candidate for the *provisional transport operator*; it does not itself provide support labels, physical rigidity, or calibrated posteriors.

The nearest implementable research target is therefore: *test whether a fully defined colored-density score plus sparse Stein interaction can produce high-precision moving seeds, under whole-sequence temporal and rigidity inference, without false motion from RGB-D jitter or occlusion.* If yes, pass the support probabilities and selected moving seeds to the existing rigid-recovery stage; jointly optimized rigid/joint models are then used downstream to compare static, multiple dynamic, and unexplained hypotheses for every unresolved point. All unproved assumptions and calibration requirements above must be settled before claiming definitive probabilistic performance.

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